

AWS Cloud Computing & IAM — Concept Notes

A simple guide to Cloud Computing, AWS Infrastructure, and IAM

1. What is Cloud Computing?

Cloud computing means using computing resources — servers, storage, databases, networking, software — over the internet ('the cloud') instead of owning and maintaining physical hardware. You rent what you need from a provider like AWS, and pay only for what you use.

Key benefits:

- **Pay-as-you-go** — no upfront hardware cost, pay for actual usage
- **Scalability** — increase or decrease resources on demand
- **Elasticity** — automatically adjust to traffic spikes
- **Global reach** — deploy applications closer to users worldwide
- **No maintenance burden** — the provider manages physical infrastructure

2. Cloud Service Models

These describe how much of the technology stack the cloud provider manages for you, versus how much you manage yourself.

Model	What it means	Example
IaaS (Infrastructure as a Service)	You get raw computing resources (servers, storage, networking). You manage the OS, apps, and data.	Amazon EC2, EBS
PaaS (Platform as a Service)	Provider manages the underlying OS and infrastructure. You just deploy your code.	AWS Elastic Beanstalk
SaaS (Software as a Service)	Fully built application, ready to use. You just use it — no management at all.	Gmail, Dropbox

3. Cloud Deployment Models

These describe where the cloud infrastructure is hosted and who can access it.

- **Public Cloud** — Infrastructure owned by a third party (AWS, Azure, GCP) and shared among multiple customers over the internet. Cheapest, most common.

- **Private Cloud** — Infrastructure used exclusively by one organization, either on their own premises or hosted privately. More control, more cost.
- **Hybrid Cloud** — A mix of public and private cloud, connected together, so data and apps can move between them as needed.

4. AWS Global Infrastructure

AWS infrastructure is organized geographically so applications can run close to users, with built-in redundancy.

- **Region** — A physical geographic area (e.g. Mumbai = ap-south-1, N. Virginia = us-east-1) containing multiple, isolated data center clusters.
- **Availability Zone (AZ)** — One or more discrete data centers within a Region, each with independent power, cooling, and networking. Most Regions have 3+ AZs, so if one fails, others keep running.
- **Edge Location** — Smaller sites used by Amazon CloudFront (AWS's CDN) to cache content closer to end users, reducing latency.

5. IAM (Identity and Access Management)

IAM is the AWS service that controls **who** can access your AWS account and **what** they're allowed to do. It lets you avoid sharing your root account credentials by creating separate identities with only the permissions they need.

6. IAM Users, Groups & Roles

Concept	Purpose
IAM User	Represents one person or application. Has its own long-term credentials (password / access keys).
IAM Group	A collection of users. Attach a policy once to the group, and every user in it inherits those permissions. Easier than managing permissions per-user.
IAM Role	A set of temporary permissions that can be assumed by a trusted entity — an AWS service (like EC2), an application, or a user from another AWS account. No permanent credentials; access is temporary and revocable.

Simple analogy: A User is like an employee ID card. A Group is a department (e.g. 'Finance') that all employees in it share access rules for. A Role is like a visitor badge — temporary access, given only for the task at hand, then it expires.

7. IAM Policies

A policy is a JSON document that defines permissions — what actions are Allowed or Denied on which AWS resources. Policies are attached to Users, Groups, or Roles.

- **AWS Managed Policies** — Pre-built by AWS for common use cases (e.g. AmazonS3ReadOnlyAccess).
- **Customer Managed Policies** — Created and controlled by you, for custom permission needs.
- **Inline Policies** — Attached directly to a single user/group/role, not reusable elsewhere.
- **Principle of Least Privilege** — Always grant only the minimum permissions needed to do the job, nothing more.

8. AWS Shared Responsibility Model

Security in AWS is a shared job between AWS and the customer:

AWS is responsible for (Security OF the Cloud)	Customer is responsible for (Security IN the Cloud)
Physical data centers, hardware, the global network, and virtualization infrastructure.	Configuring IAM users and permissions correctly, encrypting data, patching the OS and applications, configuring network settings (security groups), and managing their own data.

Simple way to remember it: AWS secures the cloud itself (buildings, hardware, network). You secure what you put IN the cloud (your data, your access controls, your configurations).